

Lab Assignment 3

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23CS30029

1. SQL Program

```
CREATE TABLE Topic (  
    topic_id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE University (  
    university_id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    location VARCHAR(100)  
);
```

```
CREATE TABLE Program (  
    program_id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    type VARCHAR(20) NOT NULL,  
    duration INT NOT NULL  
);
```

```
CREATE TABLE Student (  
    student_id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    date_of_birth DATE NOT NULL,  
    category VARCHAR(50),  
    skill_level VARCHAR(50),  
    country VARCHAR(50)  
);
```

```
CREATE TABLE Instructor (  
    instructor_id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,
```

```

        department VARCHAR(100)
    );

CREATE TABLE OnLineContent (
    content_id SERIAL PRIMARY KEY,
    name VARCHAR(200) NOT NULL,
    type VARCHAR(50),
    url VARCHAR(500)
);

CREATE TABLE Book (
    book_id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    author VARCHAR(100)
);

CREATE TABLE Course (
    course_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    duration INT NOT NULL,
    university_id INT NOT NULL REFERENCES University(university_id)
);

CREATE TABLE Teaches (
    instructor_id INT REFERENCES Instructor(instructor_id),
    course_id INT REFERENCES Course(course_id),
    PRIMARY KEY (instructor_id, course_id)
);

CREATE TABLE Enrollment (
    student_id INT REFERENCES Student(student_id),
    course_id INT REFERENCES Course(course_id),
    score INT CHECK (score BETWEEN 0 AND 100),
    PRIMARY KEY (student_id, course_id)
);

CREATE TABLE CourseTopic (
    course_id INT REFERENCES Course(course_id),

```

```

        topic_id INT REFERENCES Topic(topic_id),
        PRIMARY KEY (course_id, topic_id)
    );

CREATE TABLE CourseOnlineContent (
    course_id INT REFERENCES Course(course_id),
    content_id INT REFERENCES OnlineContent(content_id),
    PRIMARY KEY (course_id, content_id)
);

CREATE TABLE CourseBook (
    course_id INT REFERENCES Course(course_id),
    book_id INT REFERENCES Book(book_id),
    PRIMARY KEY (course_id, book_id)
);

CREATE TABLE CourseProgram (
    course_id INT REFERENCES Course(course_id),
    program_id INT REFERENCES Program(program_id),
    PRIMARY KEY (course_id, program_id)
);

INSERT INTO University (name, location) VALUES
('IITKGP', 'West Bengal'),
('IITG', 'Assam');

INSERT INTO Program (name, type, duration) VALUES
('AI Certificate', 'Certificate', 6),
('Data Science Diploma', 'Diploma', 12),
('CS Degree', 'Degree', 48);

INSERT INTO Topic (name) VALUES
('AI'), ('Databases'), ('ML');

INSERT INTO Book (name, author) VALUES
('Deep Learning', 'Goodfellow'),

```

```
('AI Basics', 'Russell'),  
('DB Systems', 'Silberschatz');
```

```
INSERT INTO Course (name, duration, university_id) VALUES  
('GenAI', 6, 1),  
('Intro to AI', 6, 1),  
('ML Foundations', 8, 2),  
('Database Systems', 12, 2);
```

```
INSERT INTO CourseTopic VALUES  
(1,1),(2,1),(3,1),(4,2);
```

```
INSERT INTO CourseProgram VALUES  
(1,1),(2,1),(3,2),(4,3);
```

```
INSERT INTO CourseBook VALUES  
(1,1),(2,2),(3,1),(4,3);
```

```
INSERT INTO Instructor (name, department) VALUES  
('Andrew Ng', 'AI'),  
('Dr Sharma', 'CSE'),  
('Dr Rao', 'Data Science');
```

```
INSERT INTO Teaches VALUES  
(1,1),(1,2),(2,4),(3,3);
```

```
INSERT INTO Student (name, date_of_birth, category, skill_level,  
country) VALUES  
('Amit', '2010-05-10', 'School', 'Beginner', 'India'),  
('Ravi', '1960-01-01', 'Retired', 'Beginner', 'India'),  
('John', '1999-07-12', 'UG', 'Intermediate', 'USA'),  
('Sara', '2001-09-20', 'UG', 'Advanced', 'UK');
```

```
INSERT INTO Enrollment VALUES  
(1,1,85),(2,1,78),(3,1,88),(4,1,90),  
(1,2,70),(3,2,75),  
(3,3,82),  
(2,4,65);
```

```
INSERT INTO OnlineContent (name,type,url) VALUES
('GenAI Intro Video','video','x.com/1'),
('AI Notes','pdf','x.com/2');
```

```
INSERT INTO CourseOnlineContent VALUES
(1,1),(2,2);
```

```
SELECT DISTINCT c.name
FROM Course c
JOIN CourseProgram cp ON c.course_id = cp.course_id
JOIN Program p ON p.program_id = cp.program_id
JOIN CourseTopic ct ON c.course_id = ct.course_id
JOIN Topic t ON t.topic_id = ct.topic_id
WHERE p.type = 'Certificate'
      AND t.name = 'AI'
      AND c.duration ≤ 6;
```

```
SELECT DISTINCT c.name
FROM Course c
JOIN CourseProgram cp ON c.course_id = cp.course_id
JOIN Program p ON p.program_id = cp.program_id
JOIN CourseTopic ct ON c.course_id = ct.course_id
JOIN Topic t ON t.topic_id = ct.topic_id
JOIN University u ON u.university_id = c.university_id
WHERE p.type = 'Certificate'
      AND t.name = 'AI'
      AND u.name = 'IITKGP'
      AND c.duration ≤ 6;
```

```
SELECT s.name
FROM Student s
JOIN Enrollment e ON s.student_id = e.student_id
JOIN Course c ON c.course_id = e.course_id
WHERE c.name = 'GenAI'
      AND (
```

```
        EXTRACT(YEAR FROM AGE(CURRENT_DATE, s.date_of_birth)) < 18
    OR EXTRACT(YEAR FROM AGE(CURRENT_DATE, s.date_of_birth)) > 60
);
```

```
SELECT DISTINCT s.name
FROM Student s
JOIN Enrollment e ON s.student_id = e.student_id
JOIN Course c ON c.course_id = e.course_id
JOIN CourseTopic ct ON c.course_id = ct.course_id
JOIN Topic t ON t.topic_id = ct.topic_id
JOIN University u ON u.university_id = c.university_id
WHERE s.country <> 'India'
      AND t.name = 'AI'
      AND u.name = 'IITKGP';
```

```
SELECT DISTINCT s.country
FROM Student s
JOIN Enrollment e ON s.student_id = e.student_id
JOIN Course c ON c.course_id = e.course_id
JOIN Teaches t ON t.course_id = c.course_id
JOIN Instructor i ON i.instructor_id = t.instructor_id
WHERE i.name = 'Andrew Ng';
```

```
SELECT DISTINCT i.name
FROM Instructor i
JOIN Teaches t ON t.instructor_id = i.instructor_id
JOIN Course c ON c.course_id = t.course_id
JOIN Enrollment e ON e.course_id = c.course_id
JOIN Student s ON s.student_id = e.student_id
WHERE s.country = 'India';
```

```
SELECT DISTINCT c.name
FROM Course c
JOIN Enrollment e ON c.course_id = e.course_id
WHERE e.student_id IN (
    SELECT e2.student_id
    FROM Enrollment e2
    JOIN Course g ON g.course_id = e2.course_id
```

```
WHERE g.name = 'GenAI'
);
```

```
SELECT c.name
FROM Course c
WHERE NOT EXISTS (
    SELECT *
    FROM Enrollment e1
    WHERE e1.course_id = c.course_id
    AND NOT EXISTS (
        SELECT *
        FROM Enrollment e2
        JOIN Course g ON g.course_id = e2.course_id
        WHERE e2.student_id = e1.student_id
        AND g.name = 'GenAI'
    )
);
```

```
SELECT c.name
FROM Course c
JOIN University u ON u.university_id = c.university_id
JOIN Enrollment e ON e.course_id = c.course_id
WHERE u.name = 'IITKGP'
GROUP BY c.course_id, c.name
ORDER BY COUNT(e.student_id) DESC
LIMIT 1;
```

```
SELECT s.name
FROM Student s
JOIN Enrollment e ON s.student_id = e.student_id
JOIN Course c ON c.course_id = e.course_id
JOIN CourseTopic ct ON c.course_id = ct.course_id
JOIN Topic t ON t.topic_id = ct.topic_id
WHERE s.country = 'India'
    AND t.name = 'AI'
GROUP BY s.student_id, s.name
ORDER BY AVG(e.score) DESC
LIMIT 1;
```

2. Python Program

```
import psycopg2 as ps
import pandas as pd

def fetch_results(query, conn):
    cur = conn.cursor()
    cur.execute(query)
    cols = [desc[0] for desc in cur.description]
    df = pd.DataFrame(cur.fetchall(), columns=cols)
    cur.close()
    return df

def run_query(query, conn, commit=False):
    cur = conn.cursor()
    cur.execute(query)
    res = cur.fetchall()
    print(res)
    if commit:
        conn.commit()
    cur.close()

def query_answer(question, query, conn):

    print(question)
    df = fetch_results(query, conn)
    print("Response: ")
    print(df)
    print()

def main():
    conn = ps.connect(
        dbname="23CS30029",
        user="23CS30029",
        password="23CS30029"
    )
```



```
"""
```

1. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months.

```
"""
```

```
question = "1. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months."
```

```
query = """
```

```
SELECT DISTINCT Course.name
```

```
FROM Course
```

```
JOIN CourseProgram ON Course.course_id=CourseProgram.course_id
```

```
JOIN Program ON Program.program_id=CourseProgram.program_id
```

```
JOIN CourseTopic ON Course.course_id=CourseTopic.topic_id
```

```
JOIN Topic ON Topic.topic_id=CourseTopic.topic_id
```

```
WHERE Program.type = 'Certificate'
```

```
      AND Topic.name = 'AI'
```

```
      AND Course.duration ≤ 6
```

```
"""
```

```
query_answer(question, query, conn)
```

```
"""
```

2. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months that are offered in partnership with "IITKGP".

```
"""
```

```
question = "2. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months that are offered in partnership with "IITKGP"."
```

```
query = """
```

```
SELECT DISTINCT Course.name
```

```
FROM Course
```

```
JOIN CourseProgram ON Course.course_id = CourseProgram.course_id
```

```
JOIN Program ON Program.program_id = CourseProgram.program_id
```

```
JOIN CourseTopic ON Course.course_id = CourseTopic.topic_id
```

```
JOIN Topic ON Topic.topic_id = CourseTopic.topic_id
```

```

JOIN University ON University.university_id = Course.university_id
WHERE Program.type = 'Certificate'
      AND Topic.name = 'AI'
      AND University.name = 'IITKGP'
      AND Course.duration ≤ 6

```

"""

```

query_answer(question, query, conn)

```

"""

3. Names of all students having age less than 18 years or more than 60 years who have done the course named "GenAI".

"""

```

question = "3. Names of all students having age less than 18 years
or more than 60 years who have done the course named "GenAI"."

```

```

query = """

```

```

SELECT s.name

```

```

FROM Student s

```

```

JOIN Enrollment e ON s.student_id = e.student_id

```

```

JOIN Course c ON c.course_id = e.course_id

```

```

WHERE c.name = 'GenAI'

```

```

      AND (

```

```

          EXTRACT(YEAR FROM AGE(CURRENT_DATE, s.date_of_birth)) <

```

18

```

          OR EXTRACT(YEAR FROM AGE(CURRENT_DATE, s.date_of_birth))

```

> 60

```

      );

```

"""

```

query_answer(question, query, conn)

```

"""

4. Names of all students who are not Indians and have done a course having the topic "AI" that was offered in partnership with "IITKGP".

"""

question = "4. Names of all students who are not Indians and have done a course having the topic "AI" that was offered in partnership with "IITKGP"."

```
query = ""
SELECT Student.name
FROM Student
JOIN Enrollment ON Student.student_id = Enrollment.student_id
JOIN Course ON Course.course_id = Enrollment.course_id
JOIN CourseTopic on Course.course_id = CourseTopic.course_id
JOIN Topic on Topic.topic_id = CourseTopic.topic_id
JOIN University on University.university_id = Course.university_id
WHERE Student.country <> 'India'
      AND Topic.name = 'AI'
      AND University.name = 'IITKGP'
""
query_answer(question, query, conn)
```

""

5. Names of all countries from where a student has done a course instructed by "Andrew Ng".

""

question = "5. Names of all countries from where a student has done a course instructed by "Andrew Ng"."

```
query = ""
SELECT DISTINCT Student.country
FROM Student
JOIN Enrollment ON Student.student_id = Enrollment.student_id
JOIN Course ON Course.course_id = Enrollment.course_id
JOIN Teaches ON Teaches.course_id = Course.course_id
JOIN Instructor ON Instructor.instructor_id =
Teaches.instructor_id
WHERE Instructor.name = 'Andrew Ng'
""
query_answer(question, query, conn)
```

"""

6. Names of all instructors who have taught courses where at least one student was from India.

"""

question = "6. Names of all instructors who have taught courses where at least one student was from India."

query = """

```
SELECT DISTINCT Instructor.name
FROM Instructor
JOIN Teaches ON Teaches.instructor_id = Instructor.instructor_id
JOIN Course ON Course.course_id = Teaches.course_id
JOIN Enrollment ON Enrollment.course_id = Course.course_id
JOIN Student ON Student.student_id = Enrollment.student_id
WHERE Student.country = 'India'
```

"""

query_answer(question, query, conn)

"""

7. Name of courses such that at least one student who have taken this course has also taken the course named "GenAI".

"""

question = "7. Name of courses such that at least one student who have taken this course has also taken the course named "GenAI"."

query = """

```
SELECT C.name
FROM Course C
WHERE C.course_id IN (
    SELECT E.course_id FROM Enrollment E
    WHERE E.student_id IN (
        SELECT S.student_id FROM Student S
        JOIN Enrollment ER ON S.student_id = ER.student_id
        JOIN Course CR ON ER.course_id = CR.course_id
        WHERE CR.name = 'GenAI'
    )
)
```

))

"""

```
query_answer(question, query, conn)
```

```
"""
```

8. Name of all courses such that all the students who have taken this course has taken the course named "GenAI".

```
"""
```

question = "8. Name of all courses such that all the students who have taken this course has taken the course named "GenAI"."

```
query = """
```

```
SELECT c.name
```

```
FROM Course c
```

```
WHERE NOT EXISTS (
```

```
    SELECT *
```

```
    FROM Enrollment e1
```

```
    WHERE e1.course_id = c.course_id
```

```
        AND NOT EXISTS (
```

```
            SELECT *
```

```
            FROM Enrollment e2
```

```
            JOIN Course g ON g.course_id = e2.course_id
```

```
            WHERE e2.student_id = e1.student_id
```

```
                AND g.name = 'GenAI'
```

```
        )
```

```
)
```

```
"""
```

```
query_answer(question, query, conn)
```

```
"""
```

9. Name of the most popular course (in terms of number of students) that is offered in partnership with "IITKGP".

```
"""
```

question = "9. Name of the most popular course (in terms of number of students) that is offered in partnership with "IITKGP"."

```
query = """
```

```
SELECT Course.name
```

```

FROM Course
JOIN University ON Course.university_id = University.university_id
JOIN Enrollment ON Course.course_id = Enrollment.course_id
WHERE University.name = 'IITKGP'
GROUP BY Course.course_id, Course.name
ORDER BY COUNT(Enrollment.student_id) DESC
LIMIT 1
"""
query_answer(question, query, conn)

```

```

"""
10. Name of the Indian student who has got the highest average
marks considering all course on the topic "AI".
"""

```

```

question = "10. Name of the Indian student who has got the highest
average marks considering all course on the topic "AI"."
query = """
SELECT Student.name, AVG(Enrollment.score) AS avg_score
FROM Student
JOIN Enrollment ON Student.student_id = Enrollment.student_id
JOIN Course ON Enrollment.course_id = Course.course_id
JOIN CourseTopic ON Course.course_id = CourseTopic.course_id
JOIN Topic ON CourseTopic.topic_id = Topic.topic_id
WHERE Student.country = 'India'
AND Topic.name = 'AI'
GROUP BY Student.student_id, Student.name
ORDER BY avg_score DESC
LIMIT 1
"""
query_answer(question, query, conn)

```

```

if __name__ == '__main__':
    main()

```

3. Writeup

In the previous assignment, the task was to implement the E-R diagram into a PostgreSQL database. It was then filled with data and various queries were made to get specific values from the database.

In this assignment, we are supposed to use another programming language to query our database and display the responses. I have used Python as my choice of programming language to interact with the database. I have used the psycopg2 library that implements various methods to interact with the PostgreSQL database. A single database connection is created and reused for all queries. For each query, a cursor is created, the SQL statement is executed, and the result is fetched and displayed using formatted output. Pandas DataFrame is used to display query results in tabular form for better readability.

The output of the program is given below

```
python 23CS30829A3.py
1. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months.
Response:
  name
0  GenAI

2. Names of all "certificate" courses on the topic of "AI" of duration less than or equal six months that are offered in partnership with "IITKGP".
Response:
  name
0  GenAI

3. Names of all students having age less than 18 years or more than 60 years who have done the course named "GenAI".
Response:
  name
0  Amit
1  Ravi

4. Names of all students who are not Indians and have done a course having the topic "AI" that was offered in partnership with "IITKGP".
Response:
  name
0  John
1  Sara
2  John

5. Names of all countries from where a student has done a course instructed by "Andrew Ng".
Response:
  country
0  USA
1  India
2  UK

6. Names of all instructors who have taught courses where at least one student was from India.
Response:
  name
0  Dr Sharma
1  Andrew Ng

7. Name of courses such that at least one student who have taken this course has also taken the course named "GenAI".
Response:
  name
0  GenAI
1  Intro to AI
2  ML Foundations
3  Database Systems

8. Name of all courses such that all the students who have taken this course has taken the course named "GenAI".
Response:
  name
0  Intro to AI
1  Database Systems
2  GenAI
3  ML Foundations

9. Name of the most popular course (in terms of number of students) that is offered in partnership with "IITKGP".
Response:
  name
0  GenAI

10. Name of the Indian student who has got the highest average marks considering all course on the topic "AI".
Response:
  name    avg_score
0  Ravi  78.00000000000000
```